

Elijah Young

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EDUCATION

Massachusetts Institute of Technology

Bachelor of Science in Mechanical Engineering

GPA: 4.6/5.0

Cambridge, MA

Sep. 2023 – May 2027

EXPERIENCE

Equipment Engineering Intern

TSMC

Jun. 2026 – Present

Phoenix, AZ

- Engineered manufacturing operations for advanced semiconductor diffusion equipment, focusing on Ion Implant and Laser Spike Annealing (LSA) process tools in a high-volume fab environment.
- Reduced recurring equipment alarms by 15% through root-cause analysis and corrective action on LSA systems.
- Completed preventive maintenance on 30+ production tools including source head replacements and rebuilds.

Mechanical Engineering Intern

Burns & McDonnell

May 2025 – Aug. 2025

Brea, CA

- Designed commercial aviation fueling systems in the Aviation and Federal global practice.
- Developed fuel tank floor layouts for three 44,000-barrel tanks using 2D AutoCAD and Bluebeam; coordinated cross-functionally with Civil, Electrical, and Cathodic Protection teams to deliver construction-ready drawings.
- Designed a hydrant fueling system for terminal 5 at LAX servicing 15 gates, strategically placing isolation valve pits, performing fuel flow calculations and creating P&ID drawings.

Undergraduate Researcher

MIT Space Propulsion Laboratory

Sep. 2024 – May 2025

Cambridge, MA

- Designed and fabricated a rotating ionic liquid drying stand for vacuum operation at up to 2000 rpm using SolidWorks and in-house machining.
- Explored methods for creating 100 nm porous glass for ionic liquid uptake using thermal processing, sonic leaching in 3M HCl, and SEM imaging.

Student Research Fellow

Lemelson-MIT

Jun. 2024 – Aug. 2024

Cambridge, MA

- Conducted 189 qualitative interviews with K-14 educators & leaders to identify challenges in invention education.
- Analyzed 34 transcripts using Excel and Dedoose to identify systemic barriers and surfaced the phenomenon of impostor participants.

PROJECTS

NASA Big Idea Challenge | *Systems Engineer*

Sep. 2023 – Feb. 2024

- Designed an inflatable water storage device for the lunar environment, used SolidWorks and performed thermal modeling in MATLAB to submit a team proposal to NASA
- Led research on radiation shielding using the On-Line Tool for the Assessment of Radiation In Space (OLTARIS) with emphasis on the performance of water and lunar regolith against galactic cosmic rays and solar particle events

MIT FSAE Motorsports | *Aero Mounting Subteam*

Sep. 2023 – May 2024

- Utilized CAD to successfully design fasteners that attached the electric vehicle's body panels to its chassis
- Fostered cross team collaboration with 20+ people across the MechE and Aero teams to ensure proper fastening

TECHNICAL SKILLS

Software: Python, MATLAB, SolidWorks, Microsoft Excel, AutoCAD 2D, Bluebeam Revu

Machining: Mill, Lathe, 3D Printer, Laser Cutter, Waterjet

PROGRAMS AND DISTINCTIONS

Fish & Richardson | *Program Participant - School of Fish, IP Law*

Oct. 2025 – Jan. 2026

Jane Street In Focus | *Program Participant - Strategy and Product*

Jan. 2025

Blue Origin x Tepper Space Challenge | *2nd Place*

Oct. 2024 – Nov. 2025

Undergraduate Practice Opportunities Program (UPOP) | *Program Participant*

Sep. 2024 – May 2025

The Standard | *Professional Development for Men of Color*

Sep. 2023 – Present